Introduction to Embodied AI

Lecture 3: Machine Perception: Intro to Computer Vision

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Recap - What is Embodied AI?

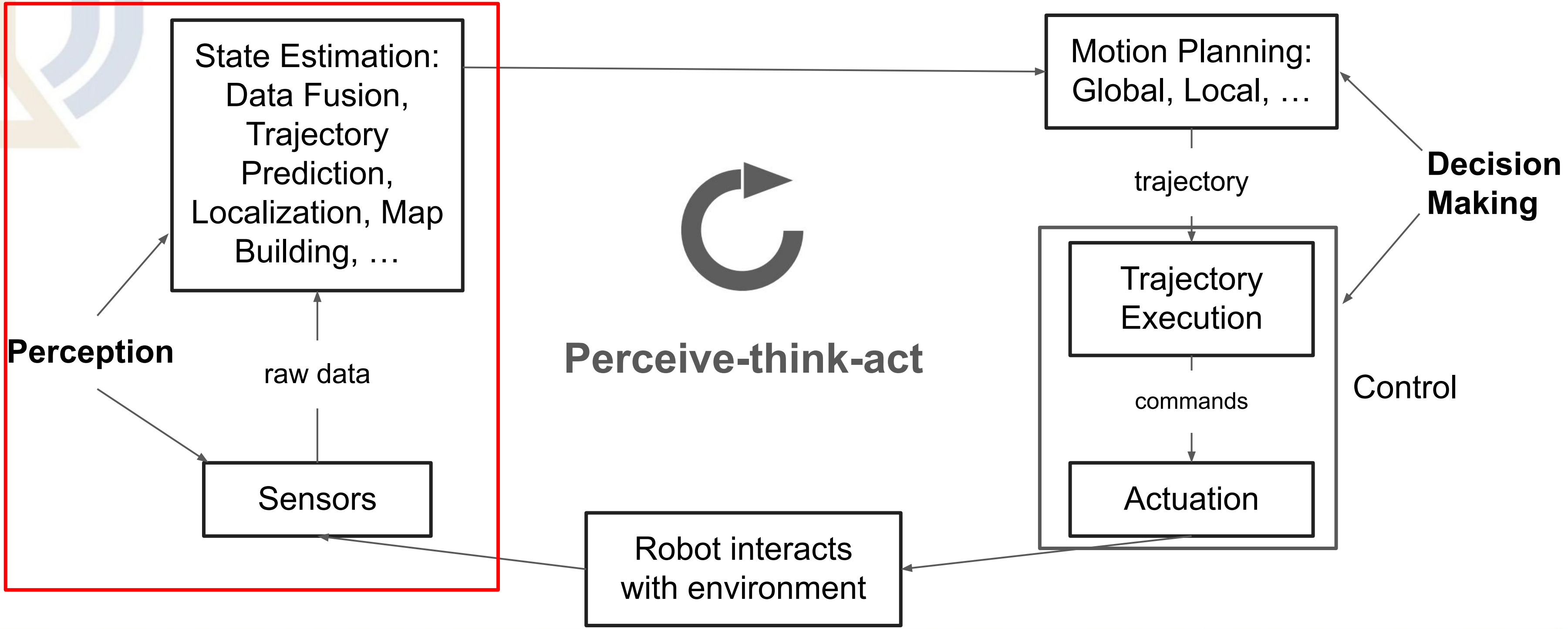
- Embodied AI is the intersection of **machine learning and robotics**
- The study of **training optimal solutions using sensory data** for robots to act and achieve goals
- **Perception** and decision making are the key

SideNote: How to Start Surveying A Research Direction?

- General idea
  - Identify key datasets, check highly-cited/latest papers using these datasets to establish baselines
- Conferences
  - Computer Vision
    - CVPR, ICCV, ECCV, SIGGRAPH (3D computer vision, NeRF/GS)
  - Vision & Language: Above + ACM Multimedia
  - Audio-Visual: Interspeech, ICASSP, + above
  - Machine Learning: NeurIPS, ICML, ICLR, AAAI, IJCAI
  - **(Learning-based) Robotics:** ICRA, IROS, RSS, CoRL / RA-L, T-RO, Science Robotics..
  - Workshops
    - CVPR AI City Challenge, Autonomous Driving, Ego4D
    - NeurIPS ML for Autonomous Driving
- Other resources
  - Survey papers
  - Awesome lists
  - Paperwithcode

Recap - The Embodied AI Paradigm

- The Perceive-think-act circle



Agenda - Intro to Computer Vision

- Objectives
  - Understand the breath of relevant tasks (and their datasets) for Robot AI research
- The tasks
  - Image Analysis (Deep dive in L5 & L6)
    - Classification (Visual recognition)
      - A review of CNNs and Transformers (See backup slides for more)
    - Object Detection and Segmentation Project 2: Egocentric Object Detection
  - Video Understanding (Deep dive in L7 & L8)
    - Object Tracking
    - Trajectory Prediction
    - Action Recognition
    - Action Detection
  - For first-person-view
    - Egocentric Perception
  - For vehicle
    - Bird-eye-view Perception

Computer Vision - Image Analysis

- Common tasks for image analysis
  - Classification (Visual recognition)
  - Object Detection
  - Segmentation

Image Classification

- Image data
    - Each pixel
        - [0, 255]
        - 3 channel - RGB
- Given an image tensor x_i of shape [H, W, 3]
- Model outputs a label

$$\hat{y}_i \in \{1, \ldots, K\}$$

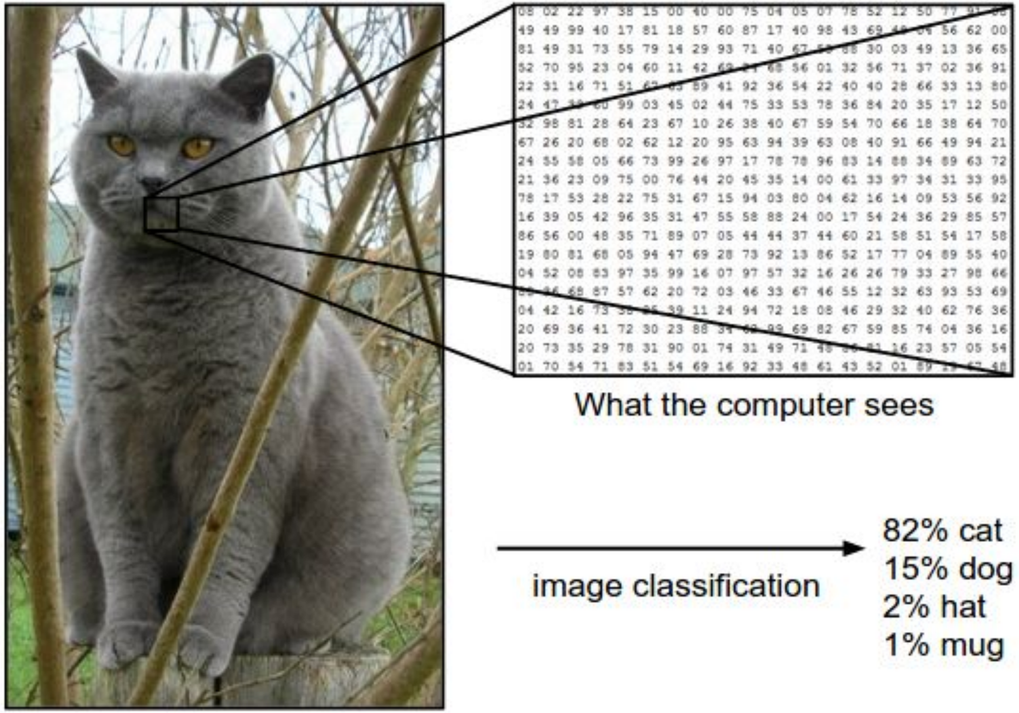


*Image from Stanford cs231n

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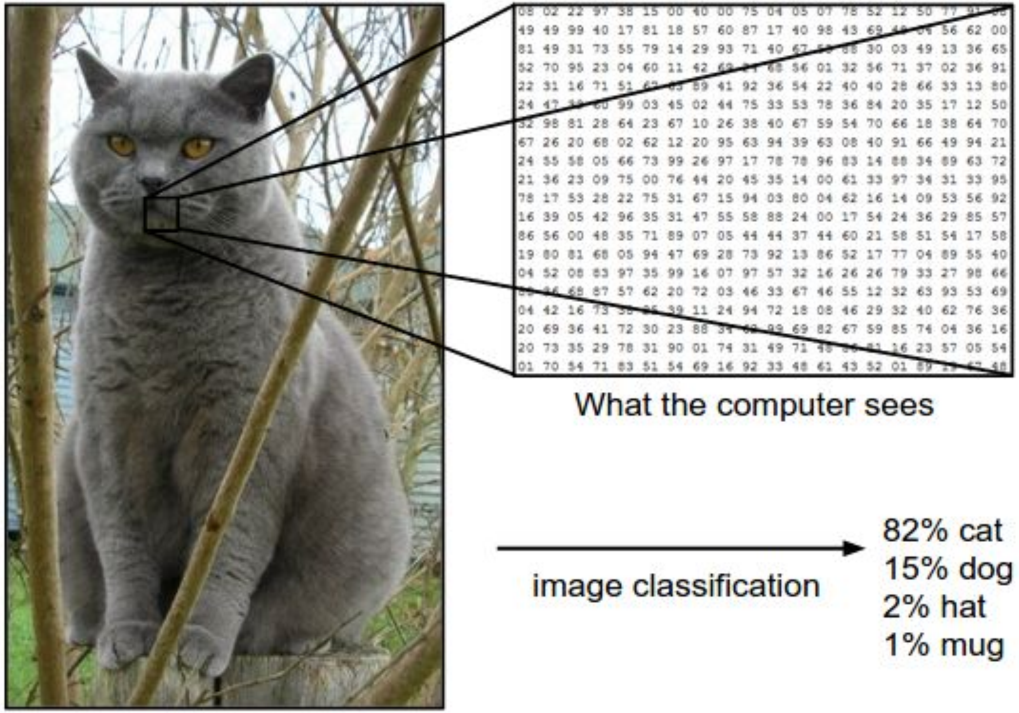
A image classification model:

$$\hat{y} = f(x; W)$$

W is the model weights, x is input image.



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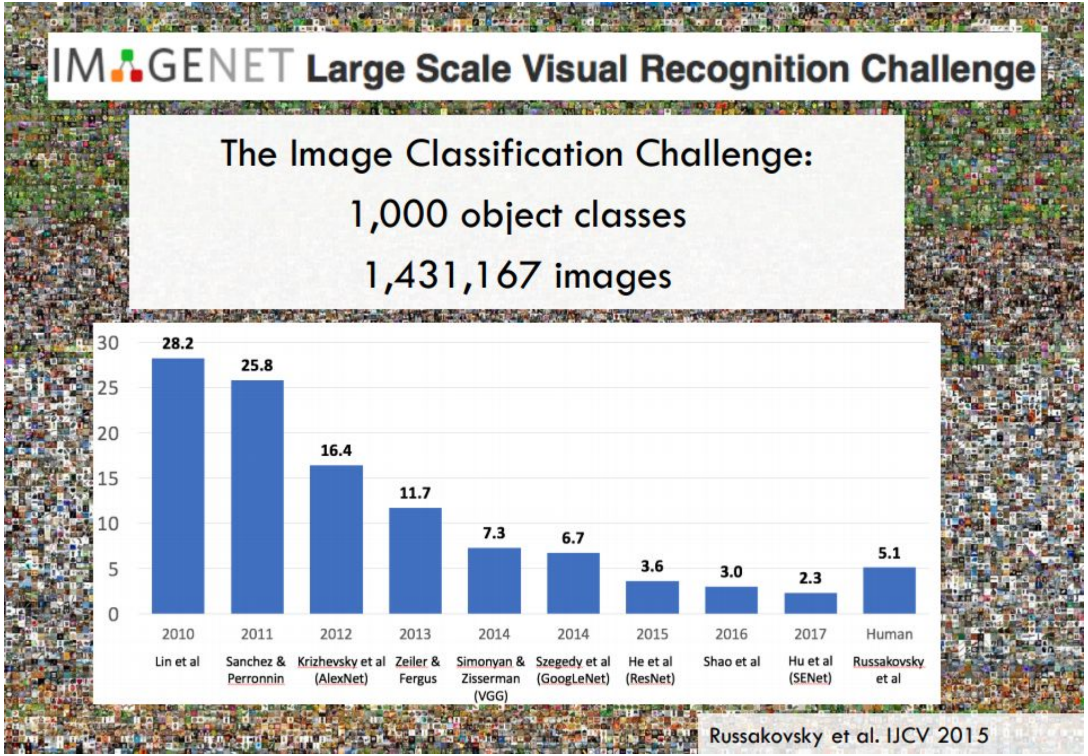
ImageNet Classification Challenge

- Challenge started in 2009
- One of the most important image dataset in computer vision



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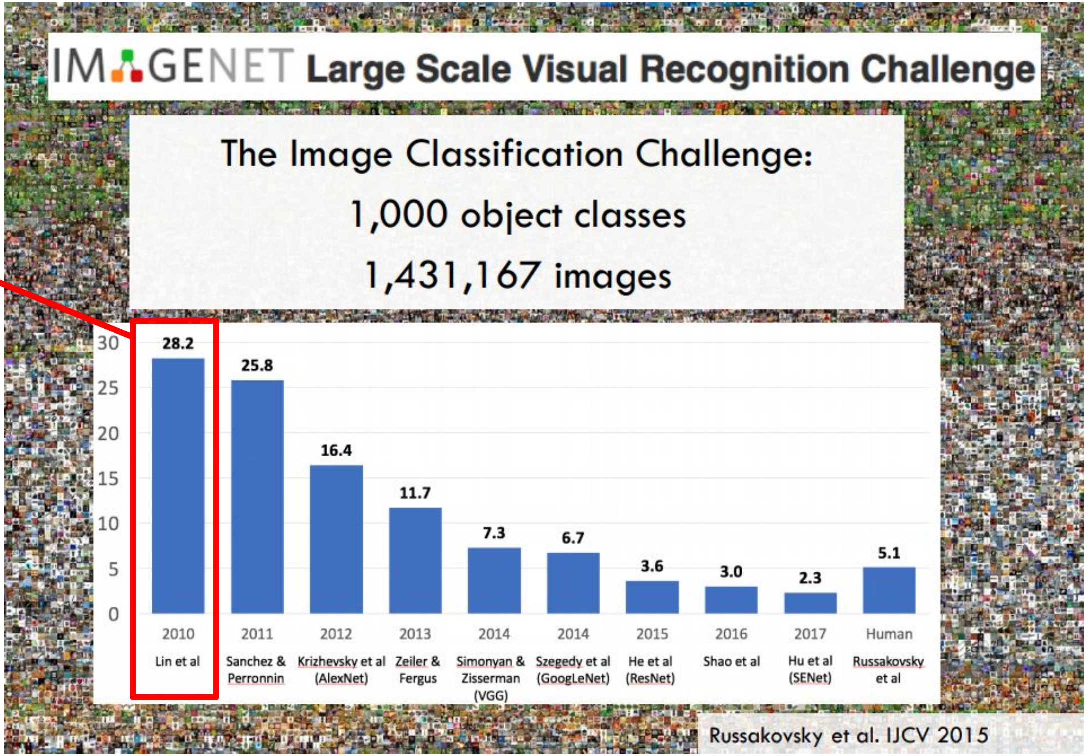
ImageNet Classification Challenge

- Local features (HOG, SIFT, etc.) -> Bag-of-Words -> SVM
-



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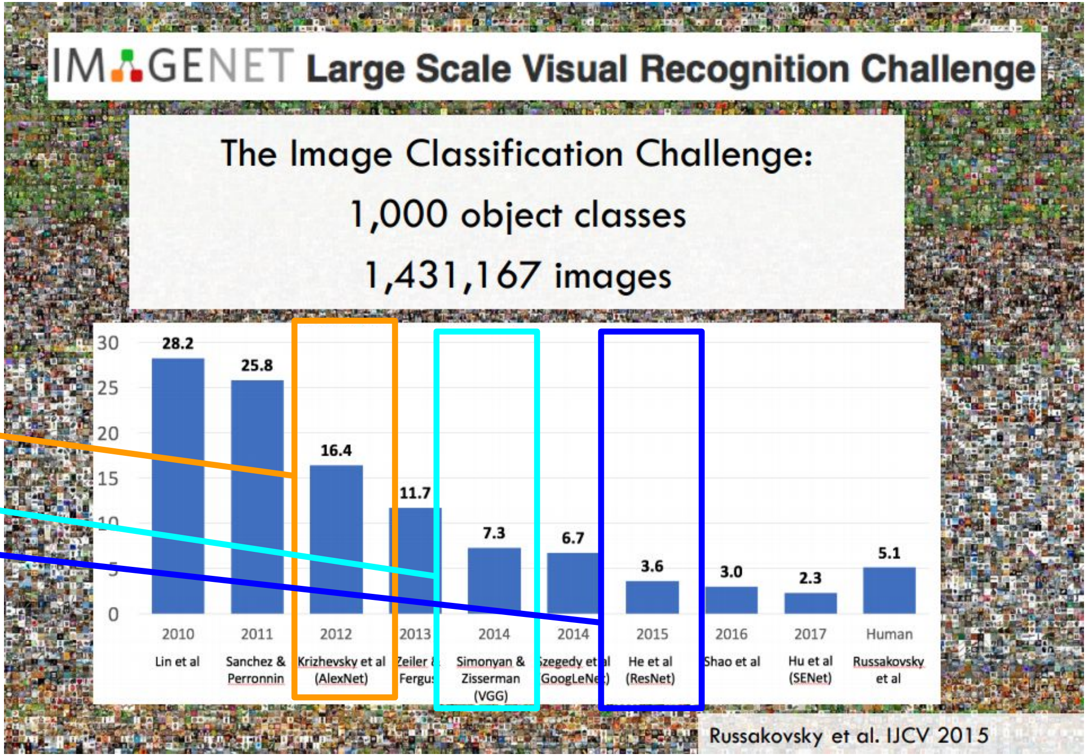
ImageNet Classification Challenge

- Local features (HOG, SIFT, etc.) -> Bag-of-Words -> SVM
- CNNs
  - **AlexNet**
  - VGGNet
  - ResNet (2016)



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Computer Vision - Image Analysis

- Common tasks for image analysis
  - Classification (Visual recognition)
    - Given an image, output probability over all N classes
    - Key datasets: MNIST, CIFAR-10/100, ImageNet-1K, ImageNet-21K, JFT-300M/3B

CIFAR-10
32x32 color images
Balanced 50k training
10k test

Image variable sizes
Usually cropped to 224x224
SOTA models use 448x448
1.4M images / 14M images

JFT image dataset:
Google's internal dataset
Noisy labels:
Image search queries +
click-through images

20 min

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    - Other special dataset: CUB-200 (Bird Recognition), Flowers-102, iNaturalist (5k species categories), Stanford Cars (196 classes)

Collectively used for fine-grained recognition
Supportive datasets for transferability

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    - Key methods
      - **CNN-based**
      - **Transformer-based**

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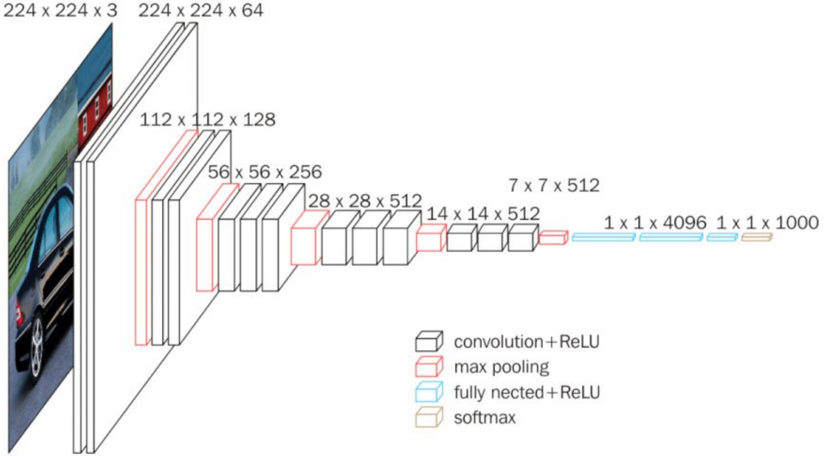
CNN-based

- VGGNet
- Inception
- ResNet
- MobileNet
- EfficientNet



Still has the advantage of inference speed

* RepVGG: Making VGG-style ConvNets Great Again, CVPR 2021

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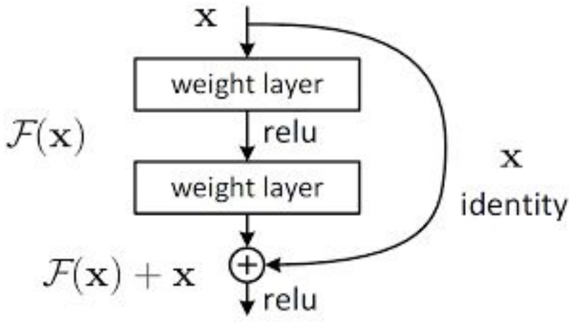

Residual connection: helps gradient flow -> solve gradient vanishing problem for deep CNNs

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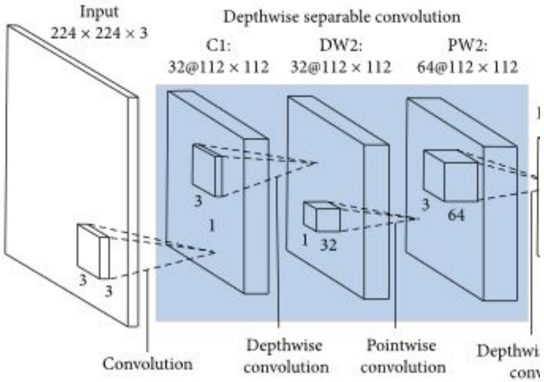

Depthwise separable convolution: conv on a single channel - much faster!

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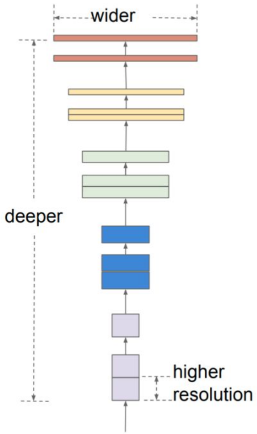

(e) compound scaling

Use AutoML to find optimal model scaling combination - width/depth/resolution

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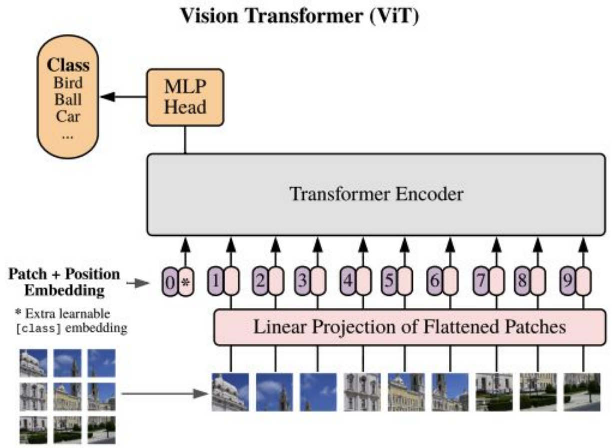
Transformer-based

- ViT (20k citation from 2020)
- Swin Transformer (10k citation from 2021)



First vision transformer that outperforms ResNet in both performance and efficiency (pretrained with JFT dataset)

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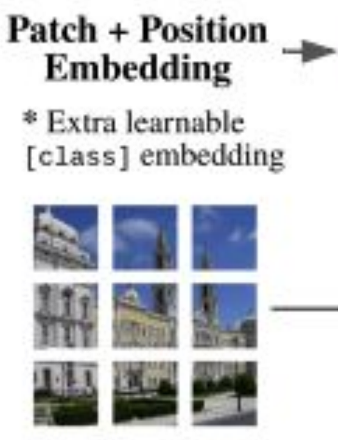

Key idea: 16x16 patch + position embedding to process original im
& self-attention/multi-head attention block

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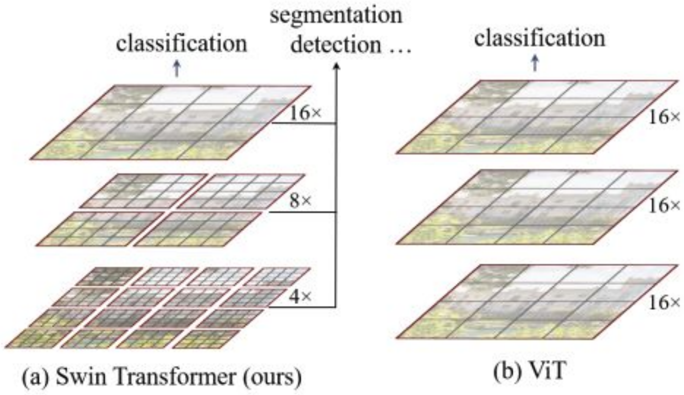
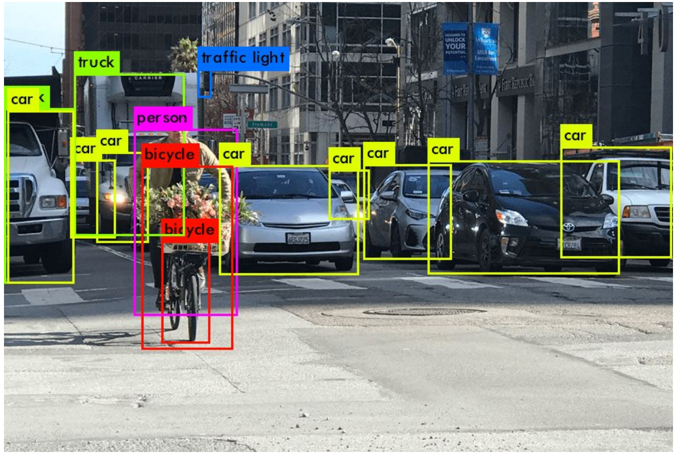


Hierarchical attention to save computation!

Computer Vision - Image Analysis

- Common tasks for image analysis
  - Detection: Given image, output bounding boxes/masks
    - Datasets: PASCAL VOC, MS COCO, Open Image



PASCAL VOC - 2012 dataset with 20 object classes.1.4K images for training

MS COCO - The go-to dataset for object detection 120K images with bounding boxes, masks and captions. 80 object classes

Computer Vision - Image Analysis

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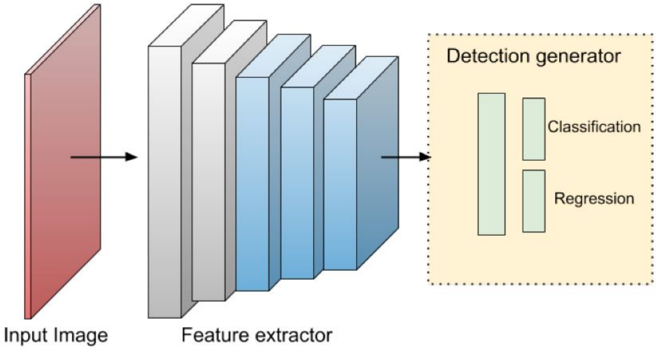
OpenImage - 9 million images (v6), partially labeled, 9600 object classes

Computer Vision - Image Analysis

- Common tasks for image analysis
  - Detection: Given image, output bounding boxes/masks
    - Datasets: PASCAL VOC, MS COCO, Open Image
    - Methods
      - Two-stage
      - One-stage



(b) One-stage RetinaNet

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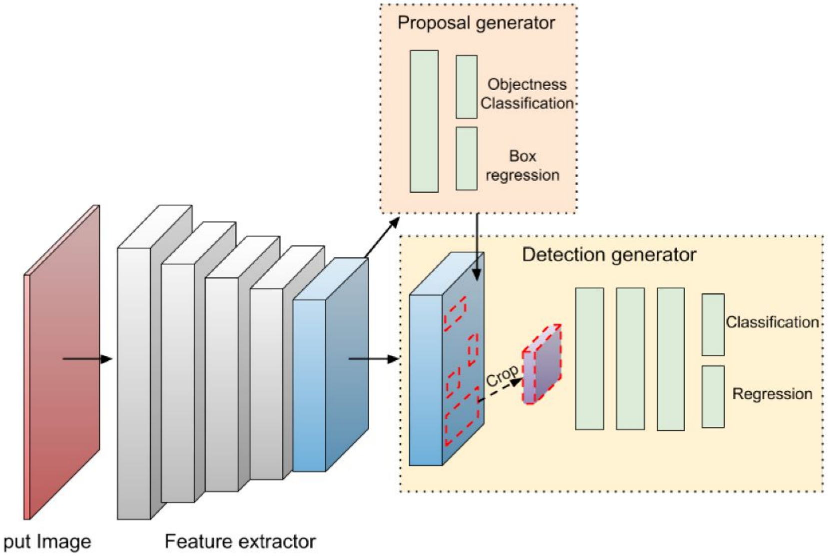
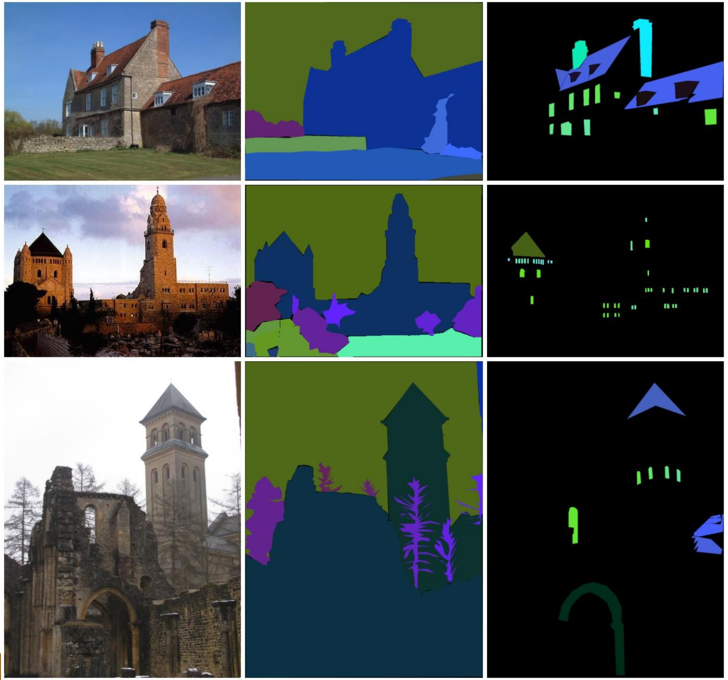


(a) Two-stage Faster R-CNN

* We will cover this in future lectures!

Computer Vision - Image Analysis

- Common tasks for image analysis
  - Segmentation: Given image, classify every pixels
    - Cityscape, ADE20K
    - Notable methods: DeepLab



ADE20K: 20K images labeled with 150 semantic classes

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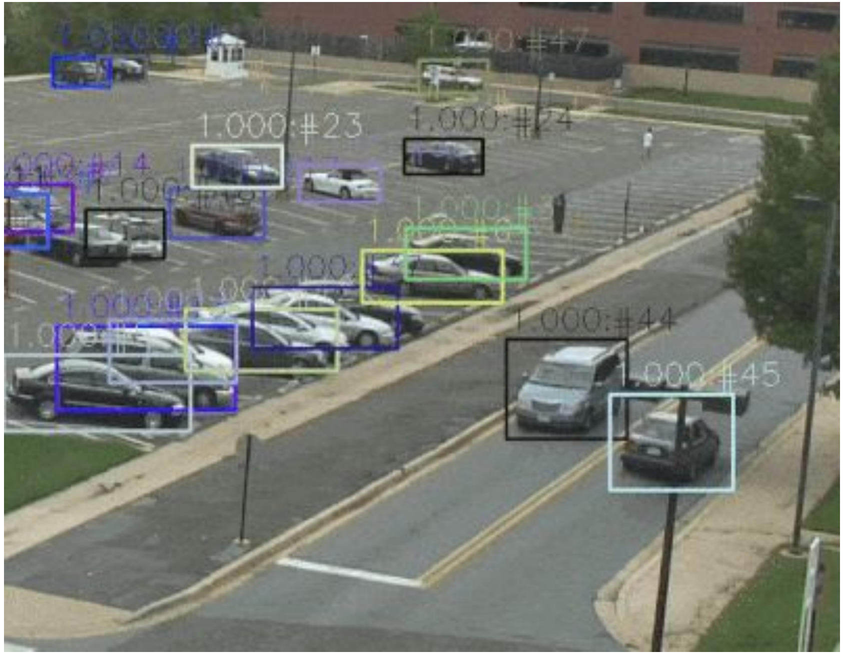
  - **Video Understanding**
    - **Object Tracking**
    - **Trajectory Prediction**
    - **Action Recognition**
    - **Action Detection**
  - For person-view
    - Egocentric Perception
  - For vehicle
    - Bird-eye-view Perception
  - Other unrelated vision tasks
    - Visual Question Answering
    - Image and Video Synthesis
    - Audio-Visual Recognition..

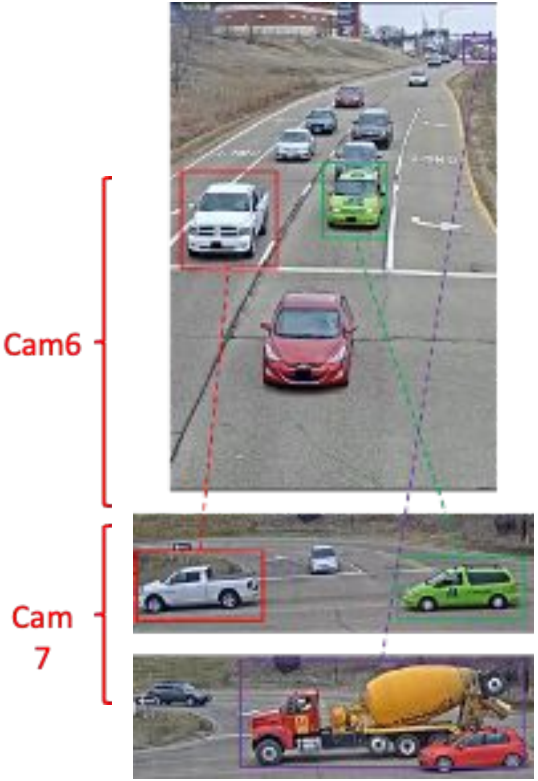
Computer Vision - Video Understanding

- Common tasks
  - Object Tracking: Need to assign “ID” for each object



Computer Vision - Video Understanding

- City-scale Vehicle Tracking and ReID
  - Multi-target multi-camera (MTMC) tracking aims to track the vehicles over large areas within multiple camera networks.
  - Different from classical multiple object tracking (MOT) which only focuses on tracking objects within a single camera, MTMC needs to resort to multiple cameras.
  - Moreover, the characteristics of moving vehicles bring unique challenges for multi-camera vehicle tracking.
  - Hosted by AI City Challenge @CVPR



Trajectory Prediction

- Problem
  - Future Person Prediction in Videos
    - Trajectory prediction / activity prediction
    - Could be used to improve traffic safety and smart robot assistants
    - Formal problem description
      - Given $X_i = (x_i^t, y_i^t), t = 1, \ldots, t_{obs}$
      - Target $Y_i = (x_i^t, y_i^t), t = t_{obs} + 1, \ldots, t_{pred}$

Trajectory Prediction

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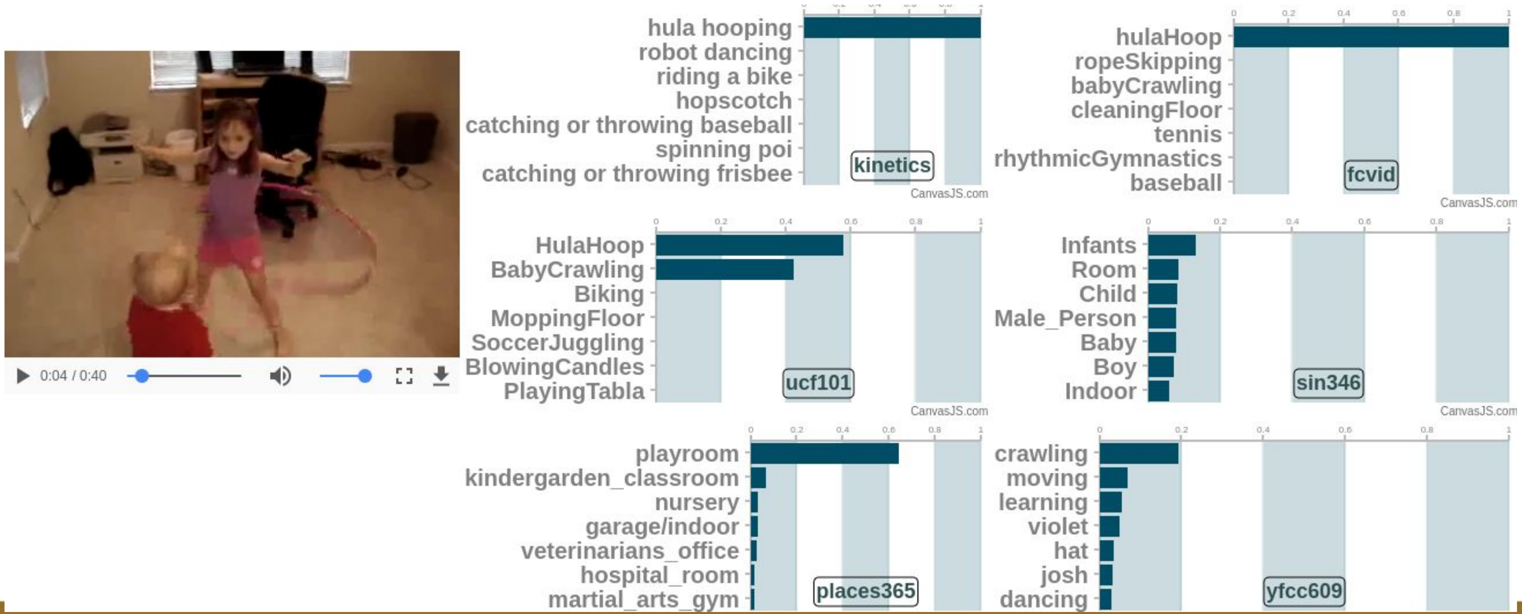
Model output examples

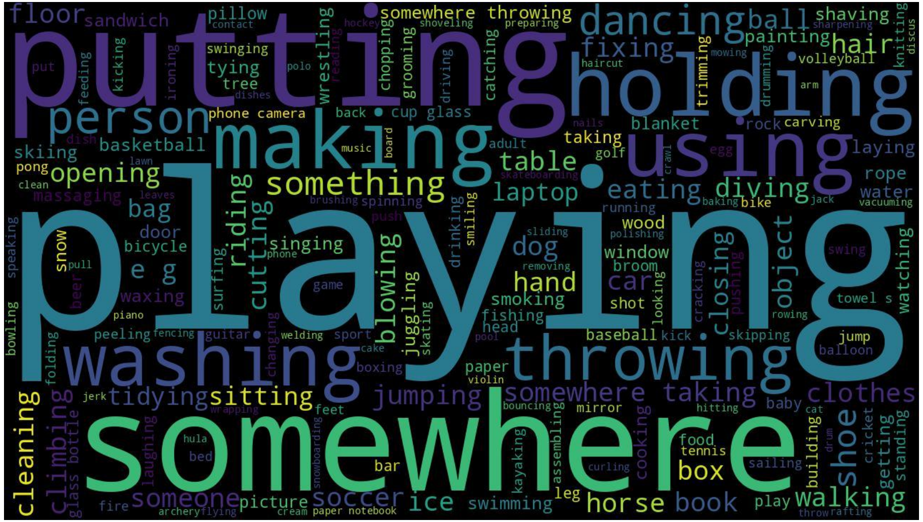
Video Classification / Action Recognition

- Given a video, output one class label



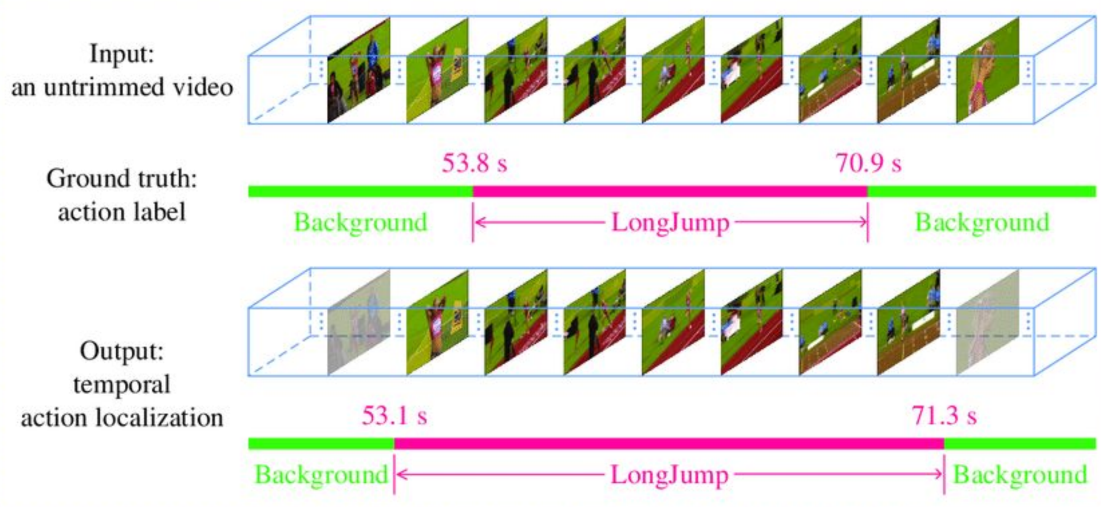
Video Classification / Action Recognition

- Common tasks
  - Action Recognition (Video Classification)
    - Datasets: UCF-101, Kinetics-400/600/700, Moments-in-Time, ActivityNet
    - Methods: I3D, SlowFast, Transformers



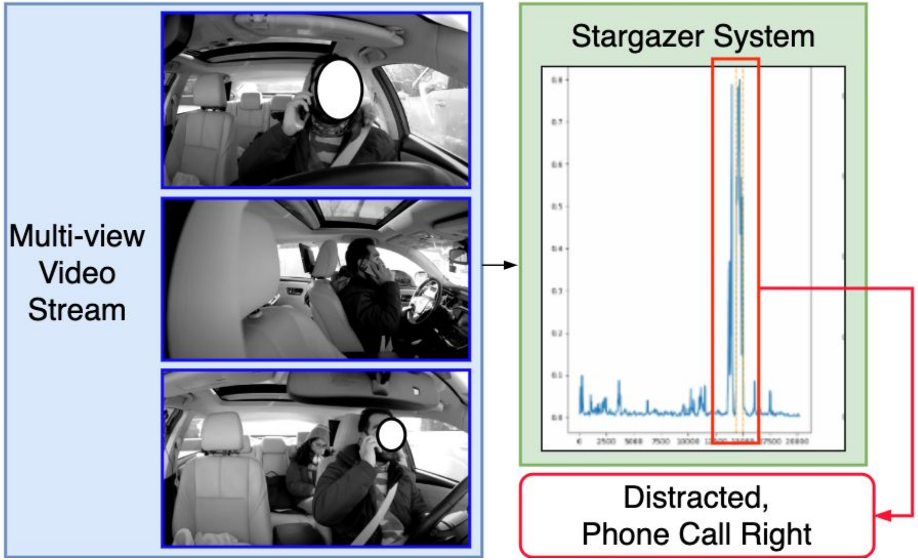
Temporal Action Localization

- Common tasks
  - Temporal Action Localization
    - Given (untrimmed) video, localize action's start and end time
    - Datasets: THUMOS, ActivityNet



Temporal Action Localization - Driver Action Localization

- Problem
  - AI City Challenge @CVPR
  - Temporally detect driver distracted actions in long videos
- Method
  - Efficient PPL with Multi-scale Vision Transformers
  - Large-scale video-text pretraining + large-scale cross-dataset robust training



Liang, Junwei, et al. "Stargazer: A transformer-based driver action detection system for intelligent transportation." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2022.

Action Detection

- Common tasks
  - Action Detection: **Given video, output bounding boxes**
    - Datasets: AVA, VIRAT, MEVA (surveillance videos)
    - Methods: See NIST TRECVID ActEV challenges

Action Detection

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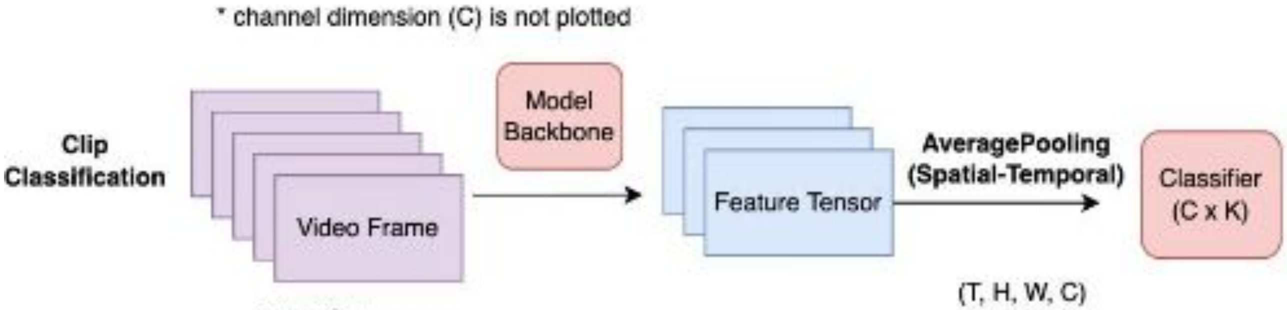
  - Comparing action recognition and action detection

Action Recognition Training



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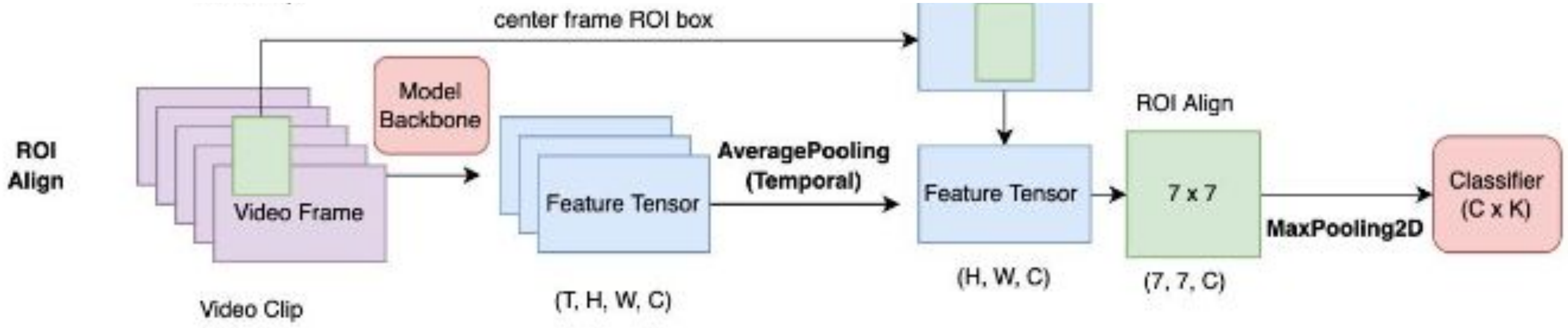
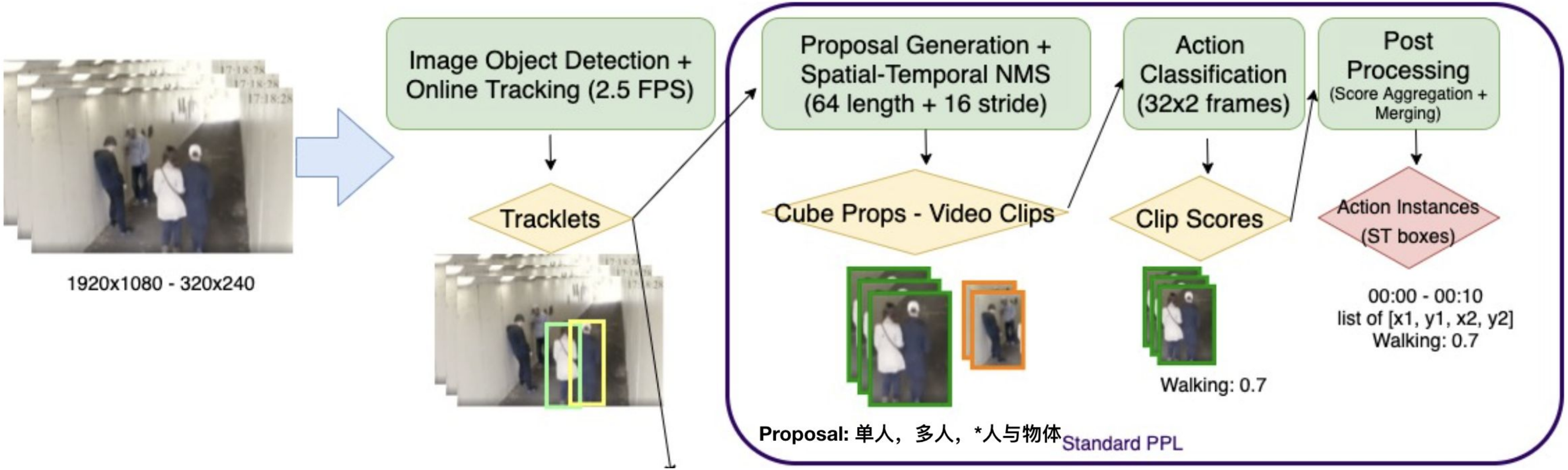
Action Detection Training



Action Detection

- Common tasks
  - Action Detection in real-world application (inference)

Inference



Computer Vision - Video Understanding

- Common tasks
  - Object Tracking: Need to assign “ID” for each object
  - Trajectory Prediction
  - Action Recognition (Video Classification)
    - UCF-101, Kinetics-400/600/700, Moments-in-Time, ActivityNet
    - Methods: I3D, SlowFast, Transformers
  - Temporal action localization: Given video, localize action’s start and end time
    - Datasets: THUMOS, ActivityNet
  - Action Detection: Given video, output bounding boxes
    - Datasets: AVA, VIRAT, MEVA (surveillance videos)
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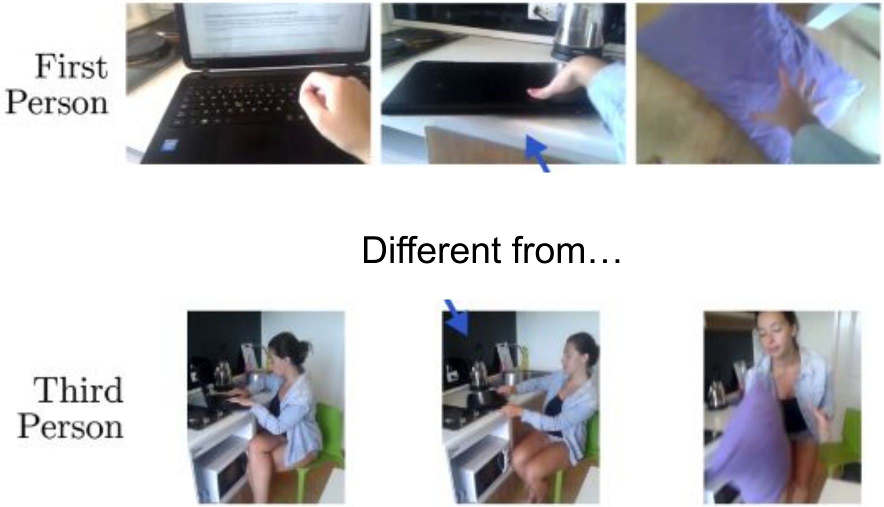
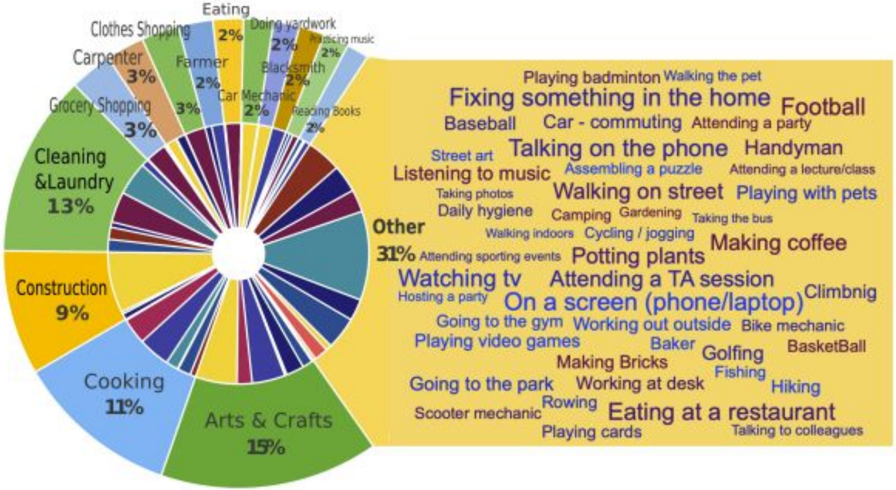
Egocentric Perception

- Or perspective view perception / First-person view perception
  - For robot vision and augmented reality



Egocentric Perception

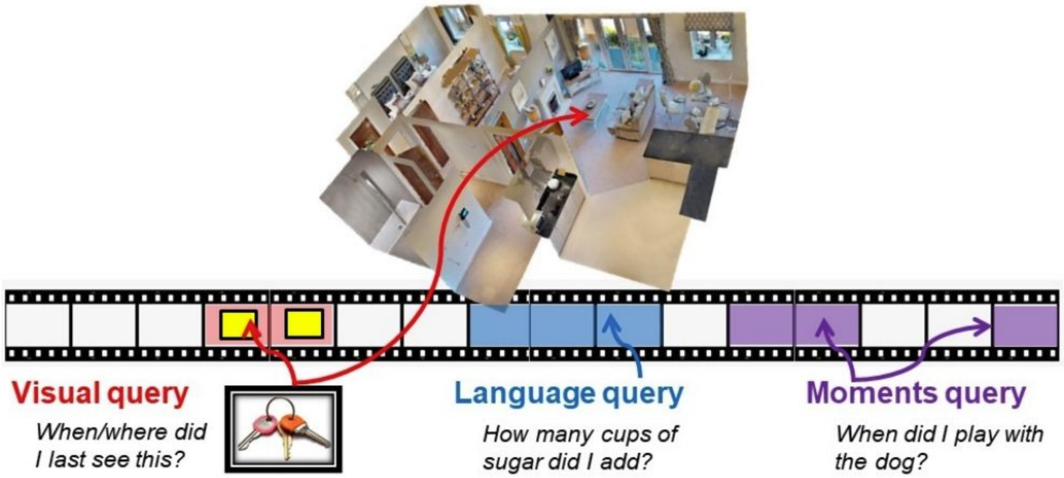
- New Dataset - Ego4D
  - 3000+ hours of ego-centric videos (head-mounted go-pros, etc.) over 74 locations in the world
  - Daily activities:



[1] Ego4D: Around the World in 3,000 Hours of Egocentric Video. CVPR 2022
[2] https://ego4d-data.org/

Egocentric Perception

- New Dataset - Ego4D
  - 3000+ hours of ego-centric videos (head-mounted go-pros, etc.) over 74 locations in the world
  - Task: Episodic Memory queries



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Egocentric Perception

- New Dataset - Ego4D
  - 3000+ hours of ego-centric videos (head-mounted go-pros, etc.) over 74 locations in the world
  - Task: Hands and Objects interaction
    - For robot demonstration learning!

State-change: Plant removed from ground

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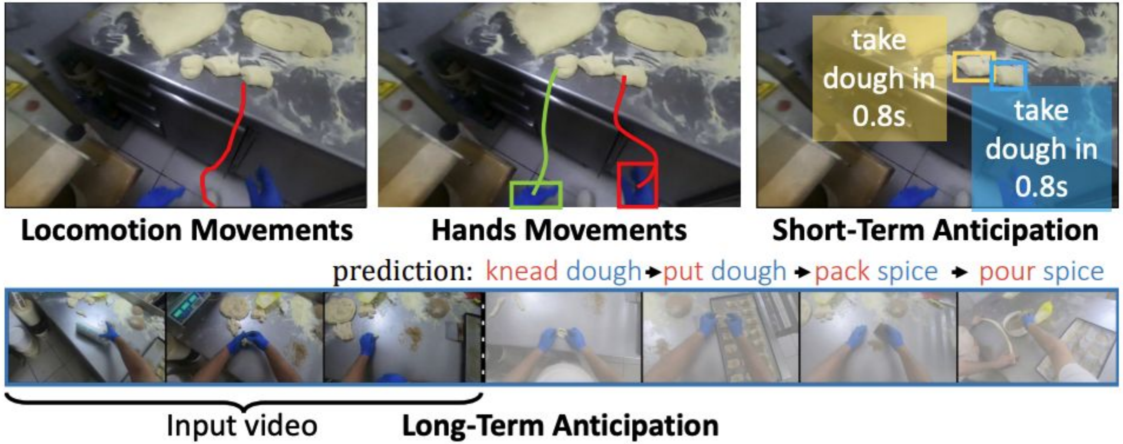
Egocentric Perception

- New Dataset - Ego4D
  - 3000+ hours of ego-centric videos (head-mounted go-pros, etc.) over 74 locations in the world
  - Task: Forecasting
    - Predict what the person will do, their hand movements



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Egocentric Perception

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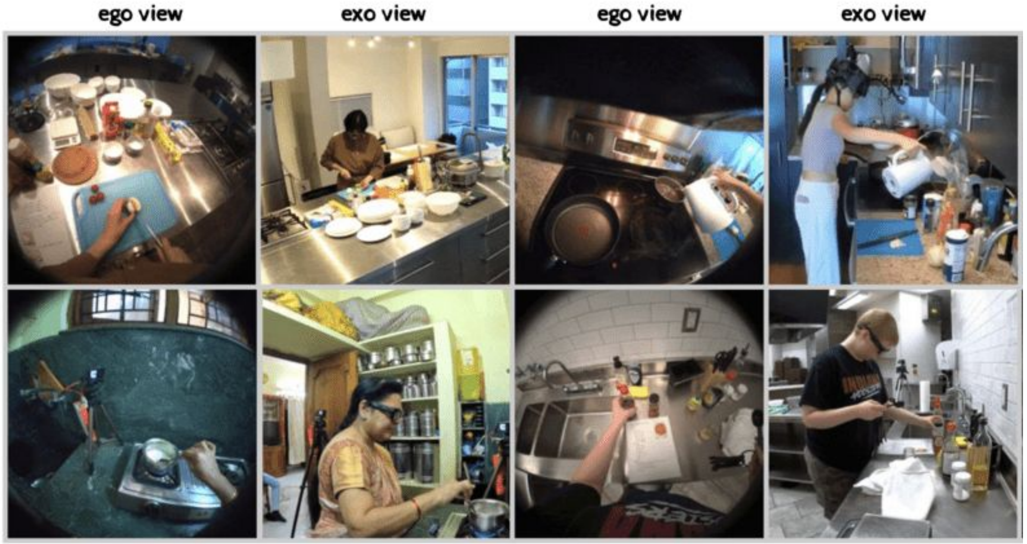
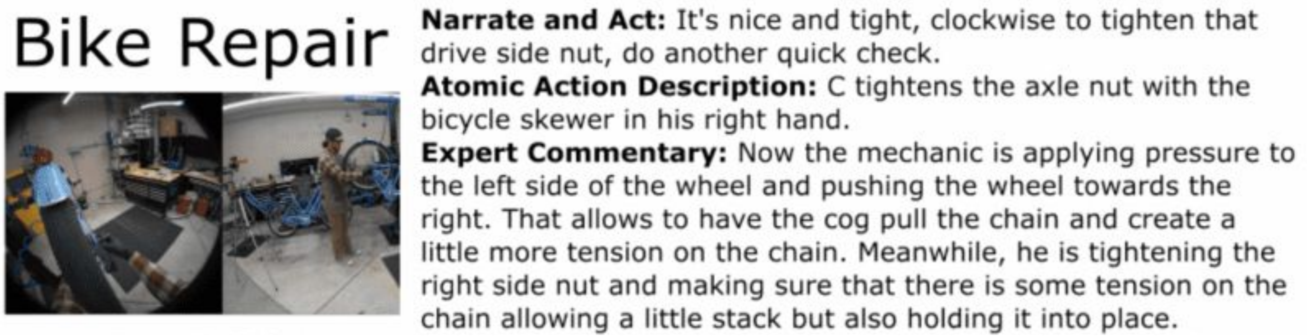
  - First-person + third person video data



[1] Ego-Exo4D: Understanding Skilled Human Activity from First- and Third-Person Perspectives. Dec 2023
[2] https://ego-exo4d-data.org/

Egocentric Perception

- New Dataset - Ego-Exo4D
  - First-person + third person video data
  - Train a pretrain model for how to accomplish certain tasks from human demonstrations



[1] Ego-Exo4D: Understanding Skilled Human Activity from First- and Third-Person Perspectives. Dec 2023
[2] https://ego-exo4d-data.org/

100 min

Agenda - Intro to Computer Vision

- Objectives
  - Understand the breath of relevant tasks (and their datasets) for Robot AI research
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  - Image Analysis
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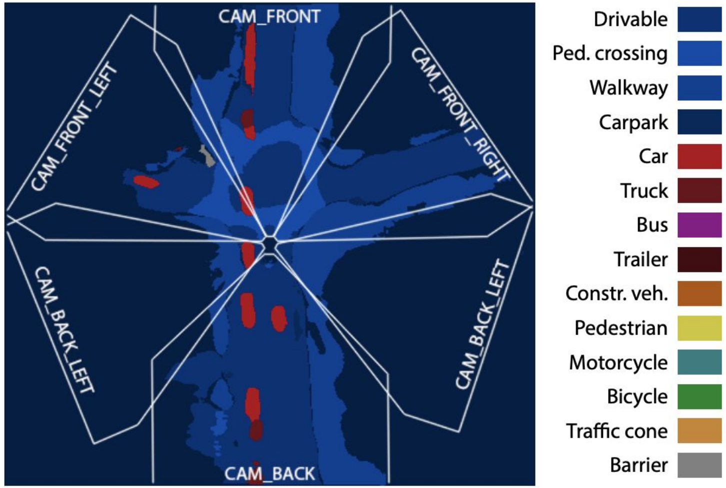
      - A review of CNNs and Transformers
    - Object Detection
    - Segmentation
  - Video Understanding
    - Object Tracking
    - Trajectory Prediction
    - Action Recognition
    - Action Detection
  - For person-view
    - Egocentric Perception
  - **For vehicle**
    - **Bird-eye-view Perception**
  - Other unrelated vision tasks
    - Visual Question Answering
    - Image and Video Synthesis
    - Audio-Visual Recognition..

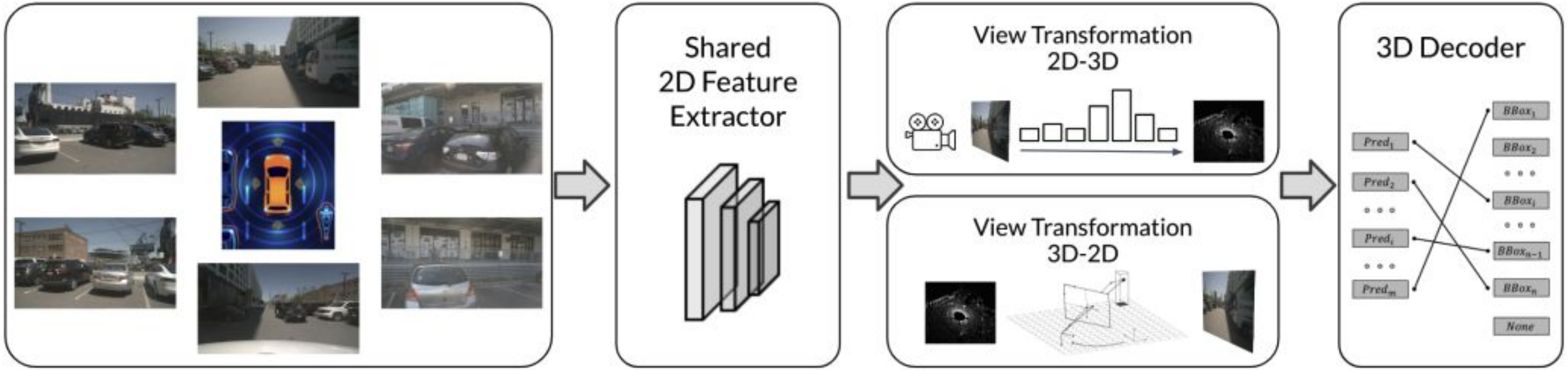
Bird-Eye-View Perception

- What is BEV perception?
  - Previously, detection/segmentation/tracking, etc., done in a front view or perspective view
    - With more cameras and sensors, need a unified view
  - From multi-camera/sensor to a top-down unified view (BEV)
  - Increasingly popular for autonomous driving algorithms



Bird-Eye-View Perception

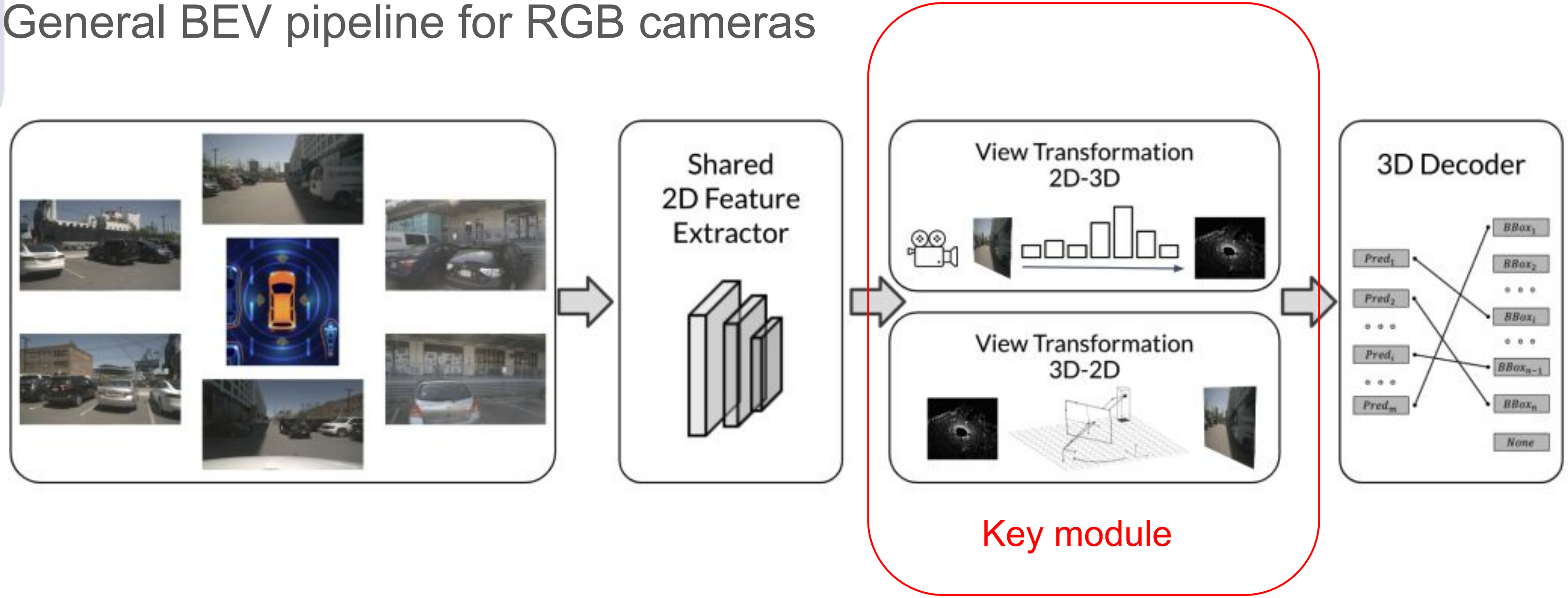
- General BEV pipeline for RGB cameras



[1] Delving into the Devils of Bird's-eye-view Perception: A Review, Evaluation and Recipe. ArXiv 2022

Bird-Eye-View Perception

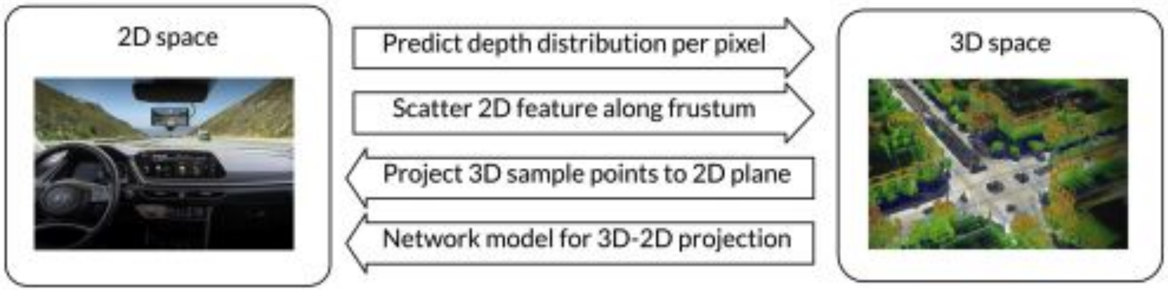
- General BEV pipeline for RGB cameras



[1] Delving into the Devils of Bird's-eye-view Perception: A Review, Evaluation and Recipe. ArXiv 2022

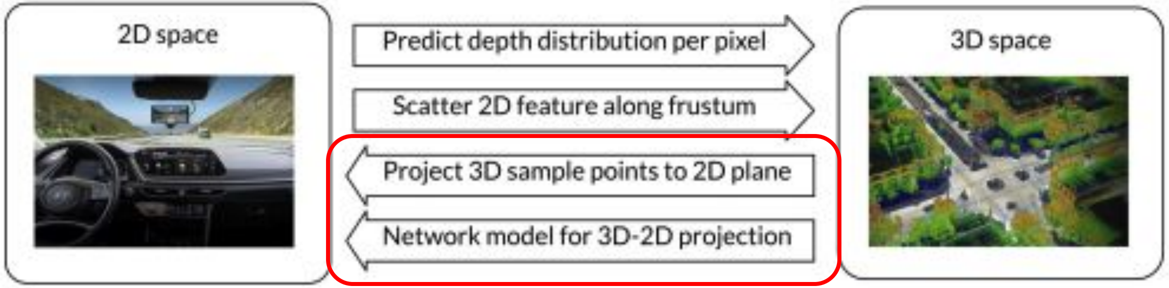
Bird-Eye-View Perception

- General BEV pipeline for RGB cameras
  - 2D-3D or 3D-2D?



Bird-Eye-View Perception

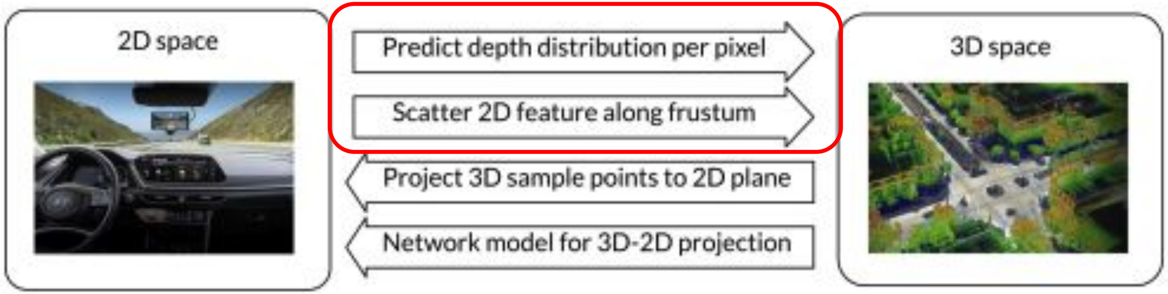
- General BEV pipeline for RGB cameras
  - 2D-3D or 3D-2D?



From 3D to 2D
1. On the BEV map, for each (3D) point, project to the camera 2D plane using camera parameters
2. Sample feature from 2D space

Bird-Eye-View Perception

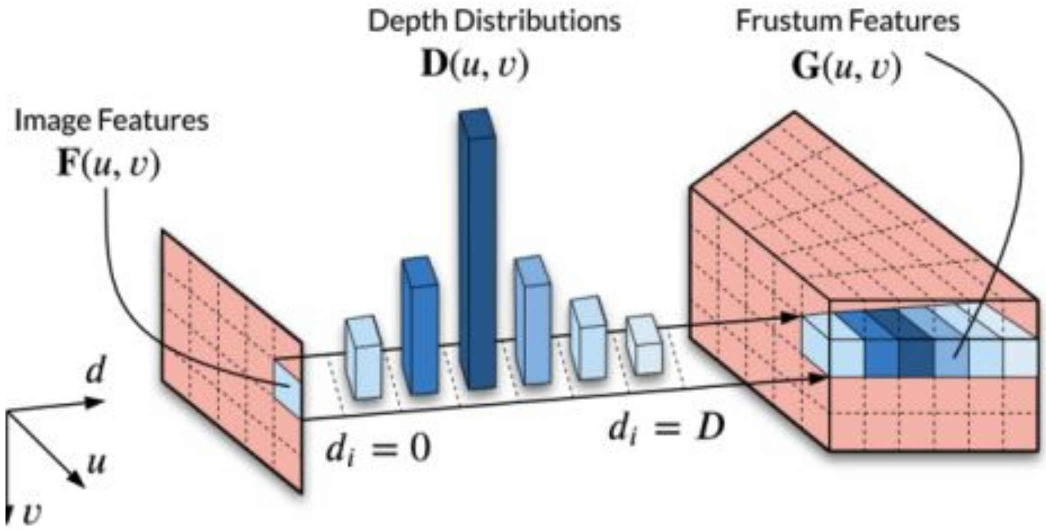
- General BEV pipeline for RGB cameras
  - 2D-3D or 3D-2D?



From 2D to 3D?
How do we get feature in 3D space directly?

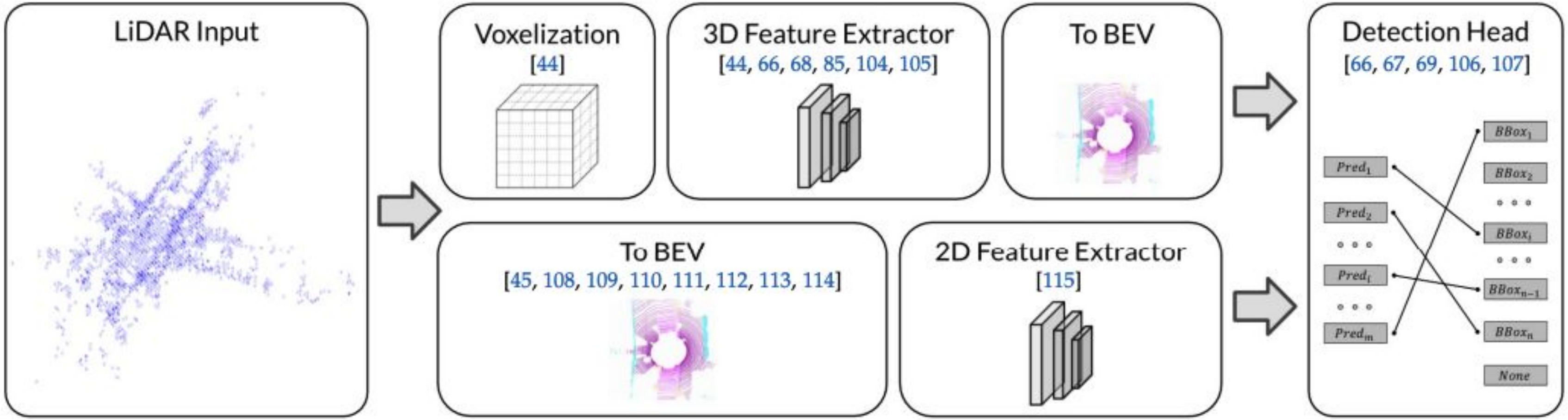
Bird-Eye-View Perception

- General BEV pipeline for RGB cameras
    - 2D-3D
        - Predict per-pixel depth as feature distribution along the projection cone
        - 2D features -> 3D voxels



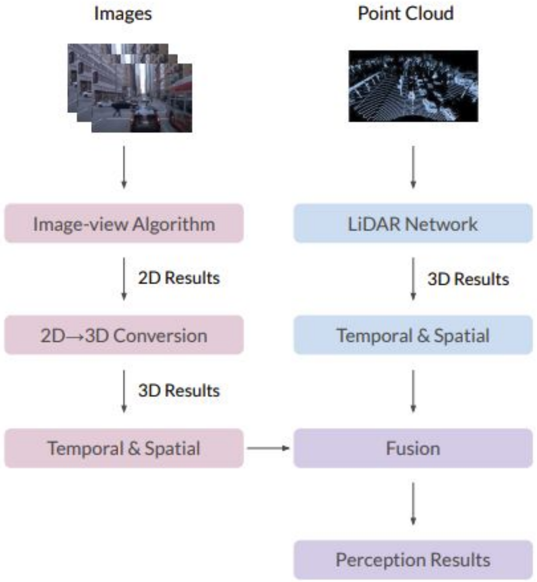
Bird-Eye-View Perception

- General BEV pipeline for LIDAR / Radar
  - LIDAR inputs are point clouds in 3D



Bird-Eye-View Perception

- Fusing RGB and LIDAR
  - Before



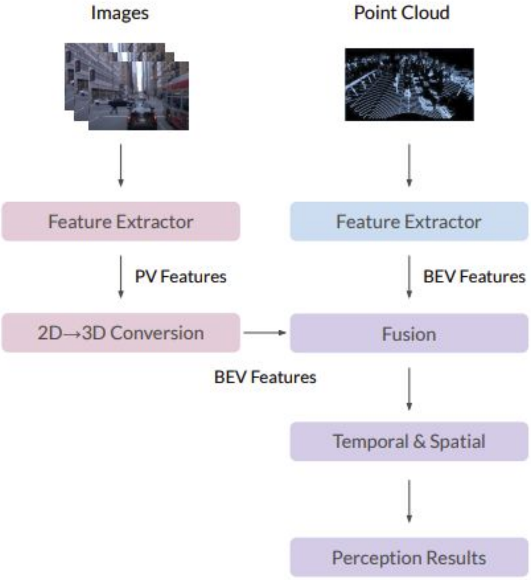
(a) Perspective view (PV) perception pipeline

Late fusion

[1] Delving into the Devils of Bird's-eye-view Perception: A Review, Evaluation and Recipe. ArXiv 2022

Bird-Eye-View Perception

- Fusing RGB and LIDAR
  - Now

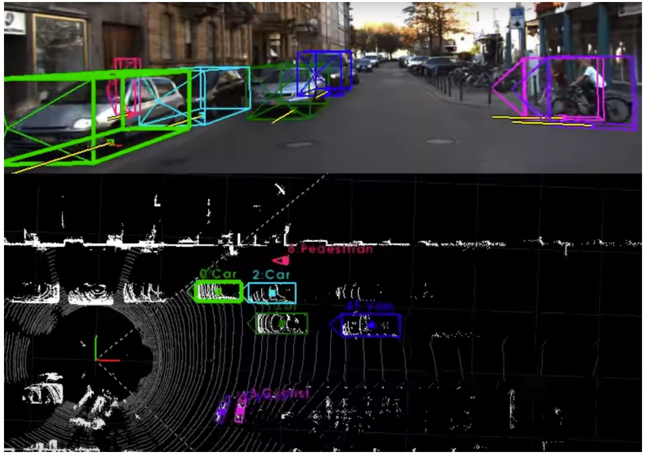


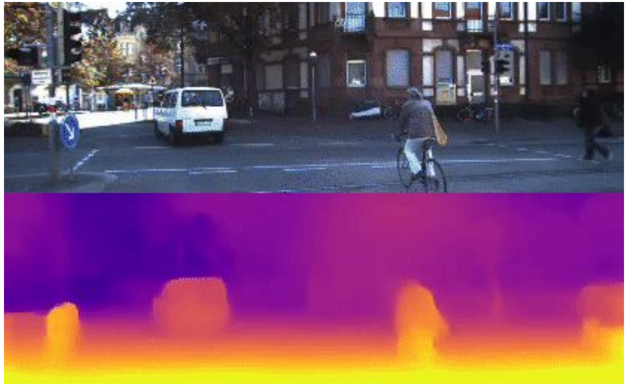
(b) BEV perception pipeline (BEV Fusion)

[1] Delving into the Devils of Bird's-eye-view Perception: A Review, Evaluation and Recipe. ArXiv 2022

Bird-Eye-View Perception

- BEV Perception datasets
  - KITTI, - RGB frames, 3D bbox, Depth*

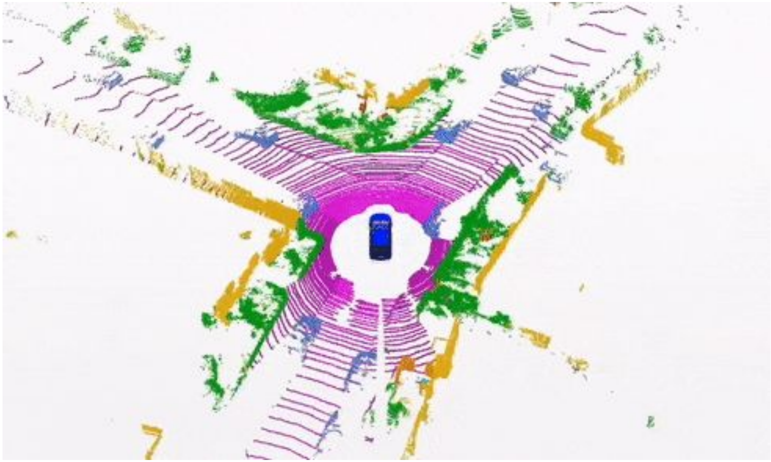




[1] Geiger, Andreas, Philip Lenz, and Raquel Urtasun. "Are we ready for autonomous driving? the kitti vision benchmark suite." CVPR 2012

Bird-Eye-View Perception

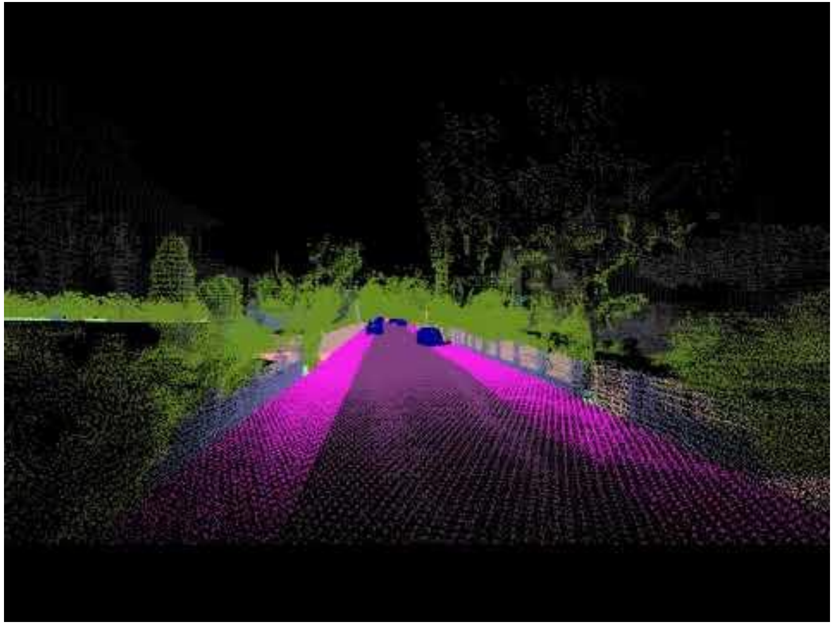
- BEV Perception datasets
  - SemanticKITTI - RGB frames, 3D bbox, 3D segmentation



[2] Behley, Jens, et al. "Semantickitti: A dataset for semantic scene understanding of lidar sequences." ICCV 2019

Bird-Eye-View Perception

- BEV Perception datasets
  - KITTI-360 - 2019



[2] Behley, Jens, et al. "Semantickitti: A dataset for semantic scene understanding of lidar sequences." ICCV 2019

Bird-Eye-View Perception

- BEV Perception datasets
  - And more
    - nuScenes
    - Waymo Open Dataset
    - Argo

Dataset	Year	City
KITTI [11]	2012	EU
Waymo [8]	2019	NA
nuScenes [7]	2019	NA/AS
Argo v1 [24]	2019	NA
Argo v2 [12]	2022	NA
ApolloScape [25]	2018	AS
OpenLane [26]	2022	NA
ONCE-3DLanes [27]	2022	AS
Lyft L5 [28]	2019	AF
A* 3D [29]	2019	AS
H3D [30]	2019	NA
SemanticKITTI [31]	2019	EU
A2D2 [32]	2020	EU
Cityscapes 3D [33]	2020	-
PandaSet [34]	2020	NA
KITTI-360 [35]	2020	EU
Cirrus [36]	2020	-
ONCE [37]	2021	AS
AIODrive [38]	2021	Sim
DeepAccident [39]	2022	Sim

[1] Delving into the Devils of Bird's-eye-view Perception: A Review, Evaluation

Bird-Eye-View Perception

- Existing BEV applications

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    - Egocentric Perception
  - For vehicle
    - Bird-eye-view Perception

Reading Material

- BEV perception survey
  - Delving into the Devils of Bird’s-eye-view Perception: A Review, Evaluation and Recipe. 2022
  - Vision-Centric BEV Perception: A Survey, 2022
- CNN/Transformer review
  - See backup slides

Quiz 1 Time!

Thank you!

CNNs and Transformers - A Quick Review

- The most simple practical classifier - Linear Classifier
  - W is model weight

$$f(x_i; W) = W x_i$$

CNNs and Transformers - A Quick Review

- The most simple practical classifier - Linear Classifier
  - W is model weight

$$f(x_i; W) = Wx_i$$

x_i [32, 32, 3]

Dog?
Cat?
Deer?
Ship?

CNNs and Transformers - A Quick Review

- The most simple practical classifier - Linear Classifier
  - W is model weight
  - Suppose we have 10 classes

[10, 3072] @ [3072]

[10]

$$f(x_i; W) = \boxed{W} x_i$$

Dog: 3.9
Cat: 1.4
Deer: -0.3
Ship: -2.2
...

x_i [32, 32, 3] -> [3072]

CNNs and Transformers - A Quick Review

- The most simple practical classifier - Linear Classifier
  - Suppose we have 10 classes
  - **N image to test**; Matrix multiplication is very fast

[10,3072] @ [3072,N]

[10,N]

$$f(x_i; W) = \boxed{W} x_i$$

Dog: 3.9
Cat: 1.4
Deer: -0.3
Ship: -2.2
...

x_i [N, 32, 32, 3] -> [3072,N]

CNNs and Transformers - A Quick Review

- Interpreting linear model weight matrix
  - Each row is a single class classifier
  - Each *ij* weight means how important is input feature dimension *j* importance to class *i*

x_i [32, 32, 3] -> [3072]

$[W_{11}, W_{12}, \ldots, W_{13071}, W_{13072}],$

$[W_{21}, W_{22}, \ldots, W_{23071}, W_{23072}],$

$\ldots$

$[W_{101}, W_{102}, \ldots, W_{103071}, W_{103072}]$

[10, 3072]

$$f(x_i; W) = \boxed{W} x_i$$

Dog: 3.9
Cat: 1.4
Deer: -0.3
Ship: -2.2
...

*Image from CMU 11777

CNNs and Transformers - A Quick Review

- How do we get W given data?
  - Stochastic gradient descent (SGD)!

x_i [32, 32, 3] -> [3072]

$$f(x_i; W) = \boxed{W} x_i$$

$$\boxed{[W_{11}, W_{12}, \ldots, W_{13071}, W_{13072}],}$$
$$[W_{21}, W_{22}, \ldots, W_{23071}, W_{23072}],$$
$$\ldots$$
$$[W_{101}, W_{102}, \ldots, W_{103071}, W_{103072}]$$

[10,3072]

Dog: 3.9
Cat: 1.4
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...

*Image from CMU 11777

CNNs and Transformers - A Quick Review

- Remember our linear classifier: $f(x_i; W) = Wx_i$
- Adding an activation function (ReLU): $f = max(0, Wx_i)$

CNNs and Transformers - A Quick Review

- Remember our linear classifier: $f(x_i; W) = Wx_i$
- Adding an activation function (ReLU): $f = max(0, Wx_i)$
- 2-layer neural network (fully-connected layer/multi-layer perceptron):
- $$f = W_2 max(0, W_1 x_i)$$

CNNs and Transformers - A Quick Review

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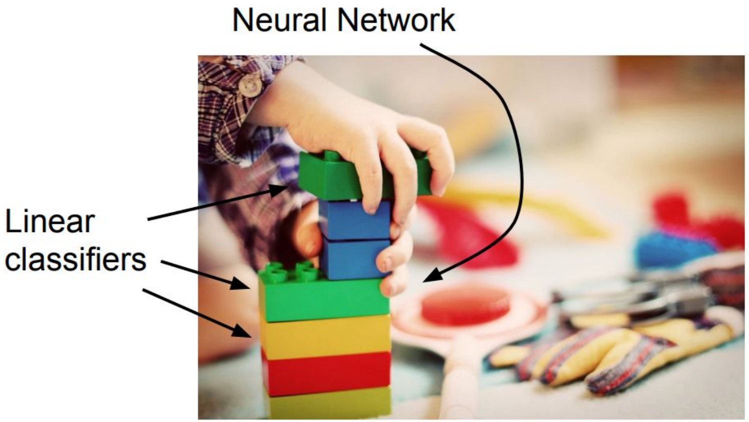


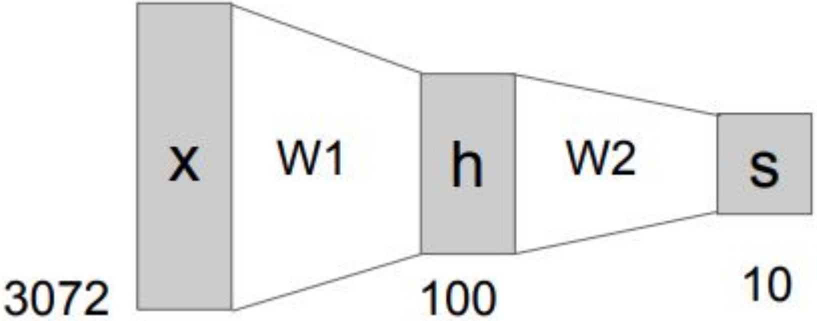
Why do we need an activation function?

CNNs and Transformers - A Quick Review

- 2-layer neural network (fully-connected layer/multi-layer perceptron) for our image classification problem:

$$f = W_2 max(0, W_1 x_i)$$

x_i [32, 32, 3] -> [3072]



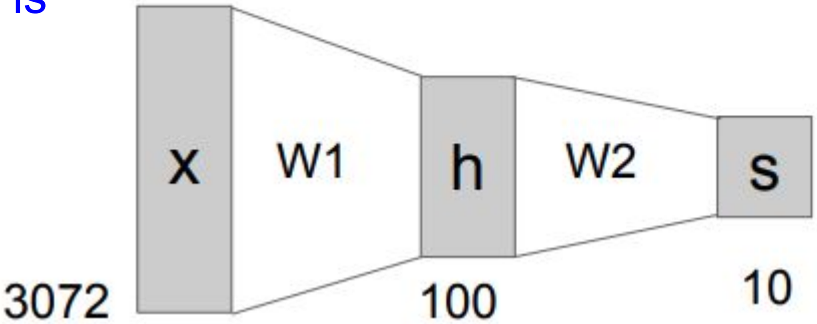
CNNs and Transformers - A Quick Review

- 2-layer neural network (fully-connected layer/multi-layer perceptron) for our image classification problem:

x_i [32, 32, 3] -> [3072]

Spatial information is lost!

$$f = W_2 max(0, W_1 x_i)$$



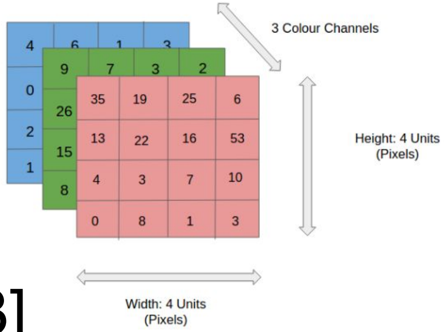
CNNs and Transformers - A Quick Review

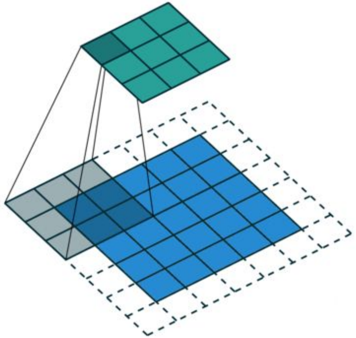
- All modern deep learning models in computer vision use CNNs
  - Even vision transformers
- Main difference to fully-connect layers
  - Input with spatial dimensions -> output with spatial dimensions

Dimension change:

[32, 32, 3] -> [32, 32, C] (stride=1) / [16, 16, C] (stride=2)

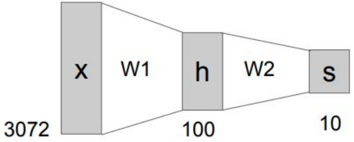
x_i [32, 32, 3]





convolutional layers

$$f = W_2 max(0, W_1 x_i)$$

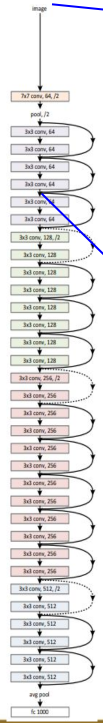


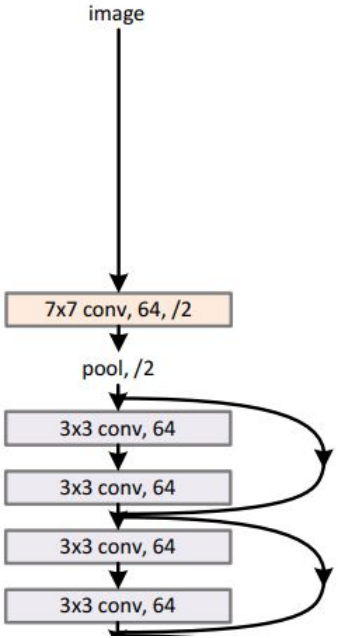
fully-connected layers

*Image from here

CNNs and Transformers - A Quick Review

- Residual neural network [CVPR 2016]



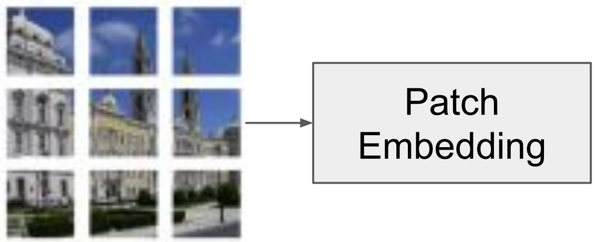


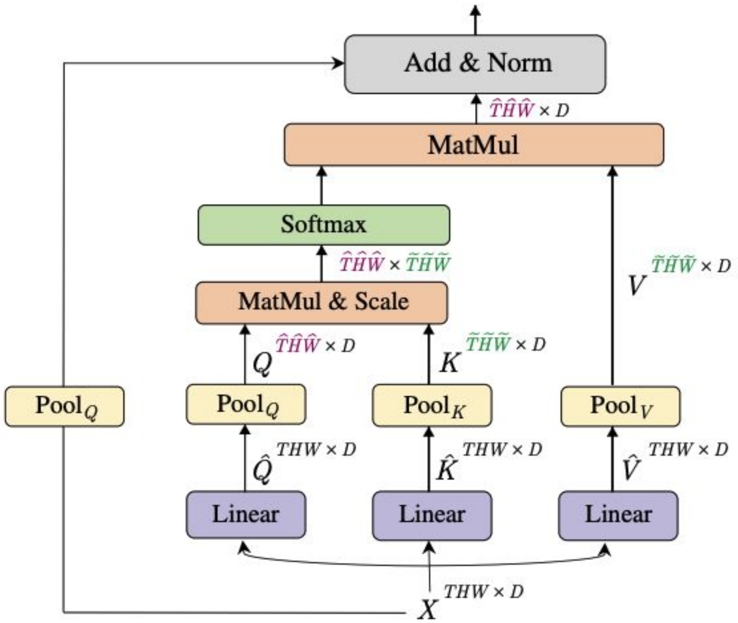
All layers are convolutional layers except that the last layer is fully-connected layer (for 1000 class image classification - ImageNet)

He, Kaiming, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. "Deep residual learning f image recognition." In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 770-778. 2016.

CNNs and Transformers - A Quick Review

- Vision Transformer
  - Two key designs: patch embedding & self-attention (multi-head attention) block





CNNs and Transformers - A Quick Review

- Vision Transformer
  - Two key designs: patch embedding & self-attention (multi-head attention) block

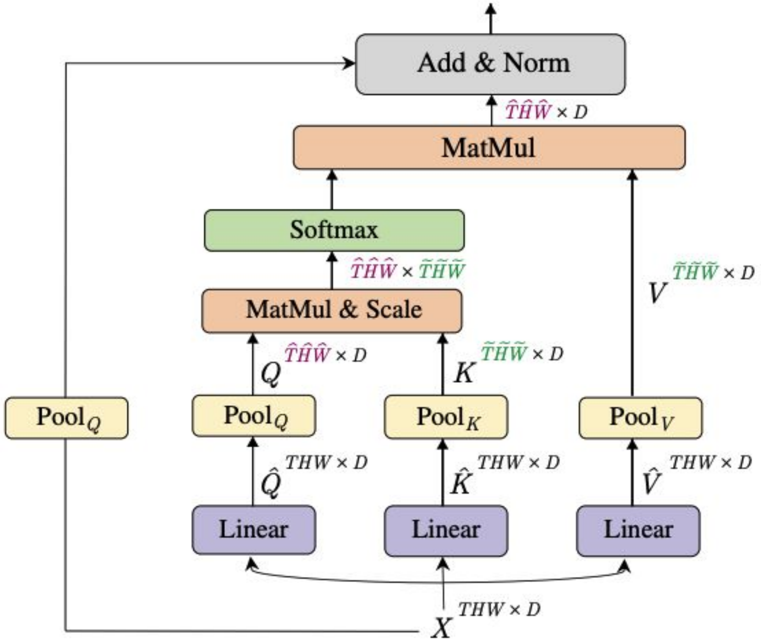
Patch Embedding

A.k.a tokenization
14 x 14 x 768

Height x Width x 3

CNNs and Transformers - A Quick Review

- Vision Transformer
  - Two key designs: patch embedding & self-attention (multi-head attention) block
    - Multi-head attention block
    - What does “attention” mean?



CNNs and Transformers - A Quick Review

- Vision Transformer
  - Two key designs: patch embedding & self-attention (multi-head attention) block
    - Multi-head attention block
    - What does “self attention” mean?
      - Think of THW as word length of a sentence
      - Self attention is the importance (weight) of each word to each other
        - Weight matrix shape: [THW, THW]